

Bokkisa Srinivas Vivek

Postdoctoral Fellow | Geotechnical Engineering

School of Civil and Environmental Engineering, Georgia Institute of Technology | Atlanta, GA, USA

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Research Profile

Geotechnical engineer and computational geomechanics researcher integrating advanced soil mechanics, numerical methods, and multiscale characterization to study granular soils and mine tailings. My work links constitutive behavior and fabric to liquefaction and instability, CPT and site characterization, and engineering-scale assessment of tailings and geo-infrastructure.

Advanced Soil Mechanics

Constitutive modelling; state dependence; fabric anisotropy; liquefaction and instability

Tailings & Geo-Infrastructure

Mine-tailings behavior; tailings-storage assessment; deformation and failure analysis

Methods & tools: MPM, FEM, DEM; constitutive-model integration and calibration; X-ray micro-CT analysis; Python, Fortran, C++, MATLAB, Julia; PLAXIS, Abaqus, Anura3D, GeoTaichi, LIGGGHTS and Yade.

Computational Geomechanics

MPM, FEM and DEM; large-deformation analysis; constitutive integration and calibration

Site Characterization & Multiscale Behavior

CPT and penetration mechanics; soil-probe interaction; X-ray micro-CT and fabric characterization

Academic Appointments & Professional Experience

Postdoctoral Fellow

May 2026–Present

Georgia Institute of Technology, School of Civil and Environmental Engineering

- Conduct research on computational geomechanics, advanced constitutive modelling, and fabric-dependent behavior of granular soils.
- Develop numerical workflows that integrate experimental data analysis, constitutive-model implementation and calibration, and engineering-scale boundary-value simulations.

Geotechnical Intern - Mining

Jun. 2024–Aug. 2024

AECOM, Denver, CO, USA

- Performed large-deformation numerical analyses of a tailings-dam section to assess runout-material pressures on a proposed retaining-wall design using the Material Point Method.
- Developed a database from previous tailings investigations and integrated rapid-vane-shear data with co-located CPT data to evaluate engineering correlations for site characterization.

Education

Ph.D. in Civil Engineering

2026

Georgia Institute of Technology

Dissertation: *Multiscale Assessment of Fabric in Granular Soils*; Advisor: Dr. Jorge Macedo; GPA: 3.91/4.0

M.S. in Computational Science and Engineering

2023

Georgia Institute of Technology

GPA: 3.75/4.0

M.Phil. in Civil Engineering

2019

Hong Kong University of Science and Technology

Dissertation: *Fabric Evolution in 3D Discrete Element Modelling of Soil Liquefaction*; Advisor: Dr. Gang Wang; GPA: 4.15/4.30

B.Tech. in Civil Engineering

2017

Indian Institute of Technology Guwahati

GPA: 8.44/10

Peer-Reviewed Journal Articles

1. **Srinivas Vivek, B.**, Macedo, J., and Arduino, P. (2026). “On the integration of an ACST-based bounding surface model.” *Computers and Geotechnics*. [DOI]
2. **Srinivas Vivek, B.**, and Macedo, J. (2025). “Challenges in NorSand to model CSD stress paths and proposed modifications.” *Canadian Geotechnical Journal*. [Link]
3. **Srinivas Vivek, B.**, and Macedo, J. (2024). “Influence of anisotropic consolidation on the instability of loose granular soils under undrained and drained loading.” *Canadian Geotechnical Journal*. [Link]
4. **Srinivas Vivek, B.**, Macedo, J., and Petalas, A. L. (2023). “Fabric anisotropy effects on static liquefaction under constant shear drained loading.” *Computers and Geotechnics*. [Link]
5. **Srinivas Vivek, B.**, Macedo, J., Petalas, A. L., and Arson, C. (2022). “Assessing flow liquefaction triggering considering fabric anisotropy effects under the ACST framework.” *Computers and Geotechnics*. [Link]
6. Kumar, A., and **Srinivas, B. V.** (2017). “Easy to use empirical correlations for liquefaction and no liquefaction conditions.” *Geotechnical and Geological Engineering*, 35(4), 1383–1407. [Link]

Manuscript Under Review

1. **Srinivas Vivek, B.**, and Macedo, J. (2026). “Fabric Effects on Cone Penetration Testing: A Numerical Investigation.” *Computers and Geotechnics*. Under review.

Conference Publications

1. **Srinivas Vivek, B.**, and Macedo, J. (2025). “Challenges in NorSand for modeling stress-relief paths in tailings storage facilities.” *Tailings and Mine Waste Conference 2025*.
2. **Srinivas Vivek, B.**, and Macedo, J. (2024). “Fabric effects on the onset of static liquefaction under drained conditions: Implications for mine tailings.” *Tailings and Mine Waste Conference 2024*.
3. **Srinivas Vivek, B.**, Macedo, J., and Petalas, A. L. (2023). “On the effects of fabric on the instability onset under constant shear drained loading.” *EMI Conference 2023*.
4. **Srinivas Vivek, B.**, Macedo, J., Petalas, A. L., and Arson, C. (2022). “Effects of soil fabric on the triggering of static liquefaction: Implications for mine tailings.” *Tailings and Mine Waste Conference 2022*.
5. **Srinivas Vivek, B.**, Macedo, J., Petalas, A. L., and Arson, C. (2022). “On the effects of fabric on the instability surface of granular materials.” *EMI Conference 2022*.
6. **Srinivas Vivek, B.**, Macedo, J., Petalas, A. L., and Arson, C. (2022). “Effects of initial consolidation on the triggering of static liquefaction considering fabric effects.” *Geo-Congress 2023*. [PDF]
7. **Srinivas Vivek, B.**, Wang, G., Duruo, H., and Feng, J. (2019). “Fabric evolution in post-liquefaction and re-liquefaction of granular soils using 3D discrete element modeling.” *Proceedings of the 7th International Conference on Earthquake Geotechnical Engineering*, Rome, Italy, vol. 4, pp. 1461–1468. CRC Press.
8. **Srinivas Vivek, B.**, and Wang, G. (2019). “DEM modelling of Toyoura sand considering real particle shapes.” *23rd Annual Conference of HKSTAM and 15th Jiangsu-Hong Kong Forum on Mechanics and its Application*, Hong Kong, p. 41.
9. **Srinivas Vivek, B.**, and Mohanty, S. (2016). “1D ground response analysis of Bhubaneswar soil in India.” *6th International Conference on Recent Advances in Geotechnical Earthquake Engineering and Soil Dynamics*, New Delhi, India.

Awards & Honors

- **George F. Sowers Distinguished Graduate Student Award**, Georgia Tech, 2026 - recognized for academic performance, scholarly contributions, professional activities, teaching, and service.
- **Outstanding Geomaterials Research Award**, Georgia Tech, 2025 - recognized for research contributions, intellectual products, oral presentation, and faculty Q&A.
- **Best Poster Award**, Annual Geotechnical Student Poster Symposium, Geo-Institute & Georgia Tech, 2024.
- **Paul W. Mayne Award**, Georgia Tech, 2024 - recognized for academic performance, research and scholarly activities, and service.
- **Postgraduate Scholarship**, Hong Kong University of Science and Technology, 2017–2019.

Teaching Experience

Graduate Teaching Assistant

Jan. 2021–May 2025

Georgia Institute of Technology

- **Courses:** Introduction to Geotechnical Engineering; Data Analytics in Civil and Environmental Engineering.
- Supported laboratory demonstrations, office hours, student questions, and grading.

Graduate Teaching Assistant

Feb. 2018–May 2019

Hong Kong University of Science and Technology

Teaching Interests: Soil Mechanics; Introduction to Geotechnical Engineering; Advanced Soil Mechanics; Computational Geomechanics; Constitutive Modelling; In-Situ Testing and Site Characterization; Tailings and Mine-Waste Geomechanics; Data Analytics for Civil and Environmental Engineering.

Student Mentoring

VIP Mentor, DeepMeLT and Extreme Events Engineering Groups

Sep. 2021–May 2025

Georgia Institute of Technology

- Mentored undergraduate researchers through weekly meetings, technical discussions, and research guidance within Georgia Tech's Vertically Integrated Projects program.

Student Academic Mentor, Geosystems Engineering

Jan. 2024–Dec. 2024

Georgia Institute of Technology

- Mentored a first-year graduate student through monthly meetings focused on academic progress, professional development, and graduate-community integration.

Student Academic Mentor

Aug. 2015–Apr. 2016

Indian Institute of Technology Guwahati

Invited Seminars & Selected Professional Development

- **Invited speaker**, Barr Engineering Co., 2023.
- **Invited lecture**, “NorSand and its Visualization,” Critical State Soil Mechanics courses at Colorado School of Mines and Georgia Tech.
- **Selected participant**, ALERT Olek Zienkiewicz School on Constitutive Modelling of Geomaterials, Prague, Czech Republic, 2025.
- Participant, APRU-IRIDeS Multi-Hazards Summer School, Tohoku University, Sendai, Japan, 2018.

Academic & Professional Service

Peer Review Service

Ad hoc reviewer for *Journal of Geotechnical and Geoenvironmental Engineering* (ASCE) and *Open Geomechanics*.

Vice President, Geotechnical Society

Aug. 2023–Aug. 2024

Georgia Institute of Technology

- Managed organizational finances and helped organize academic and community events for the Geosystems graduate-student community.

General Secretary of Academics, Association of Civil Engineers

Oct. 2015–Oct. 2016

Indian Institute of Technology Guwahati

Open-Source Research & Educational Resources

- **Enhancing NorSand Model Understanding Through Visualizations** - interactive educational resource. [[Project](#)]
- **DEM Visualization Toolkit** - open-source toolkit for analysis and visualization of Discrete Element Method data. [[GitHub](#)]